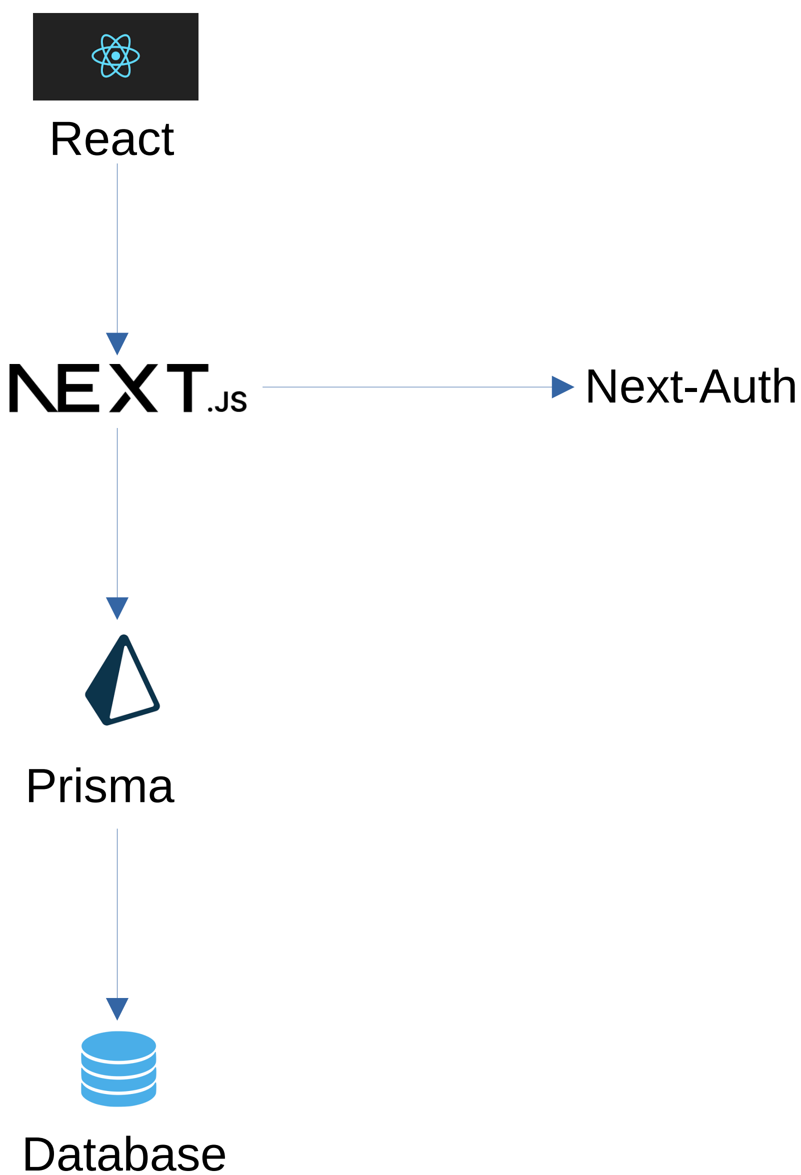
VisonAid Springboard 1.0
(Developer Documentation)

Project Description

Vision Aid partners with Eye Hospitals, Schools and NGOs serving the visually impaired in India. Springboard, specifically, is a program within Vision-Aid that is designed to improve the technical and skill development infrastructure in Blind Schools. Springboard provides grants to procure devices like Smartphones, Computers, Assistive devices etc. and sponsors mentors and trainers to impart technical, communication and skill oriented education. Vision Aid Springboard application helps Vision-Aid and the partner schools to run the Springboard program efficiently through active data organization and reporting. The application allows Vision Aid admin staff to onboard partner schools in the system and provision users of those partner schools. The staff of the partner schools can then enroll students, organize training programs, manage grants and generate reports all in the Springboard application.

Architecture Diagram



Hosting Solutions

Currently VA Springboard has 30 schools, we estimate hosting costs based on about 50 schools with 100 students per school. Each schools will have approximately 2 staff members using the Springboard application. Upper limit of user logins is 100 (approx 20 concurrent users) Upper limit database size is 10GB, considering 1MB of student data for 5000 students is 5GB + additional 5GB for training, reports etc. Expected bandwidth for 5GB student data is about 15 GB. This application can be hosted in services like AWS under less than $100 per month. Apart from AWS and GCP, here are some free hosting solutions that could be used for an application of the above size and complexity.

Vercel (https://vercel.com/)

Netlify (https://www.netlify.com/)

Application Installation

Application can be run in a docker container. Here are the steps.

Clone the repository

1. git clone https://github.gatech.edu/cs-6150-computing-for-good/va-springboard.git

2. Setup the environment variables in .env file in the application root with the database access parameters

Here is a sample file parameners.

```
DATABASE_PW=""
DATABASE_USER=""
DATABASE_NAME=""
DATABASE_HOST=""
DATABASE_PORT=""
DATABASE_URL=""
AUTH_SECRET=""
AUTH_GOOGLE_ID=""
```

3. Install node dependencies

```
npm I

4. Init the database
npm run init

5. Compile app
npm run build

6. Start Server
npm start
```

Authentication

Authentication is handled by the NextAuth library. The sign-in action calls Google's Oauth page. User logs in with their google account credentials. Googles sends back a code and Nextjs exchanges the code back for tokens access and ID tokens. NextAuth creates a session.

Database Design and Backup

Here are the database tables used in the application.

School(SCH)

ID	Int
Name	Varchar
Location	Varchar
Phone	Varchar
Email	Varchar
Tier	Varchar
Notes	Varchar

Student (STD)

ID	Int
Name	Varchar
Age	Int
Gender	Char
Visual Acuity	Varchar
Class	Int
Aadhar No	Char(12)
Phone	Varchar
Email	Varchar
GDC	Bool
BPL	Bool

Training Program (TP)

ID	Int
Name	Varchar
Description	Varchar

Topics

ID	Int
TP~ID	Int
Name	Varchar
Desc	Varchar

Activity (ACT)

ID	Int
Name	Varchar
Description	Varchar

Device (DEV)

ID	Int
Type	Varchar
Desc	Varchar
Tech param1	Varchar
Tech param2	Varchar

Enrollment

SCH~ID	Int
TP~ID	Int
Start Date	Date
End Date	Date
Sessions	int
Notes	Varchar

Beneficiary

SCH~ID	Int
STD~ID	Int
DEV~ID	Int
Issue Date	Date
Required	Bool
Existing Device Desc.	Varchar

Grant

SCH~ID	Int
MOU Date	Date
Grant	Decimal
Inf_grant	Decimal
Tp_grant	Decimal
Inf_grant_sp	Decimal
Tp_grant_sp	Decimal

Database can be backed up using Postgresql, pg_dump, here is an example usage

#pg_dump -U user -h host -d vaspringboard_db > vasp_backup.sql

Following is an example usage of using the backup to restore the database

#psql -U user -d vaspringboard_db < vasp_backup.sql

Environmental Variables

Here is the list of environmental variables used by the application for database connectivity and authentication. These are stored in a .env file in application root.

```
DATABASE_PW=""
DATABASE_USER=""
DATABASE_NAME=""
DATABASE_HOST=""
DATABASE_PORT=""
DATABASE_URL=""
AUTH_SECRET=""
AUTH_GOOGLE_ID=""
```

Partner Statement

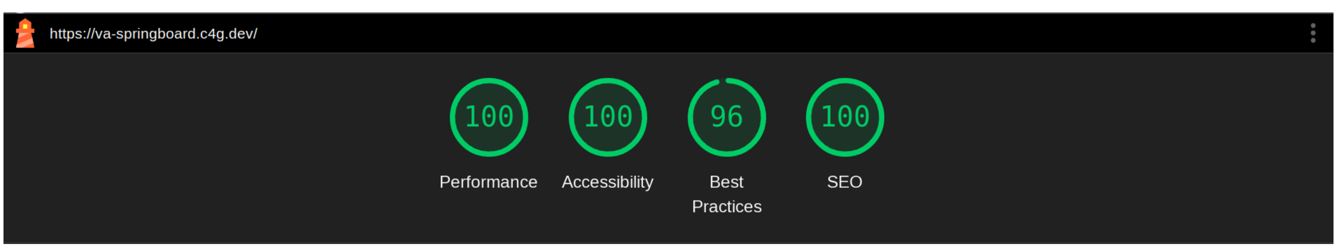
We have conducted a demo session with the partner to help them navigate through the application. Additionally, a detailed user guide has be provided to the partner to assist them using the application. As this point (as of 04/20/2025) partner understands the basics of the application, has access to the application and can navigate the application.

Installation Walk-through

Currently partners are familiarizing themselves with the application. Installation Walk-through is pending this step. Once we get time with the partner we will walk them through the application installation process.

Application UX and Metrics

Lighthouse Scores-



Form-factor Analysis (Mobile Portrait and Landscape) -

Application rendered correctly in mobile portrait and landscape mode. Pages were scroll-able both horizontally and vertically. Texts were correctly displayed without any breaks. Contents were correctly spaced out and wrapped as required by available screen real estate. Text was readable. Buttons were large enough for touch clicks. There was a issue noted with the navigation bar, where the entire navigation bar was not displayed. Mobile page screenshots are below.